WEEK 03 : PRODUCT CONTRIBUTION PHASE : DAYS 11 TO 15

Deep Learning & Attenomy Cortex Core Models

NumPy Training Loops, PyTorch Transfer Learning & Attenomy Cortex Architecture

PROGRAM & ROLE	COMPANY PRODUCT	TIME COMMITMENT	SCHEDULE & FEEDBACK
ML Engineering Intern (All-Round)	**Attenomy Cortex (Enterprise AI)**	**15-20 Hours/Week (Part-Time Unpaid)**	**Submit by Friday (11:59 PM) | Weekend Feedback**

01 Company Product Mandate : Attenomy Cortex

PRODUCT PHASE

Having completed the foundational trial practice phase in Weeks 1 and 2, you now transition directly into contributing to our company product: **Attenomy Cortex**. Attenomy Cortex is our enterprise autonomous agent orchestration and cognitive knowledge engine, powering zero-hallucination hybrid RAG, cognitive document extraction, and sub-50ms conversational agent pipelines. In Week 3, you master deep learning mechanics from first principles and fine-tune specialized neural architectures for Attenomy Cortex.

02 Gradient Foundations & NumPy Backpropagation

DAYS 11-12

```
# Numerical gradient verification formula (relative error must be below 1e-5)
def check_numerical_gradient(f, w, grad_analytical, eps=1e-7):
    grad_num = np.zeros_like(w)
    for idx in np.ndindex(w.shape):
        orig = w[idx]; w[idx] = orig + eps; l_plus = f(w)
        w[idx] = orig - eps; l_minus = f(w); w[idx] = orig
        grad_num[idx] = (l_plus - l_minus) / (2 * eps)
    return np.linalg.norm(grad_num - grad_analytical) / (np.linalg.norm(grad_num) +
np.linalg.norm(grad_analytical))
```

TASK 3.1 Neural Networks from the Ground Up (NumPy to PyTorch)

Days 11-12 : Mon-Tue

Construct a 2-layer Multi-Layer Perceptron (MLP) from scratch using only NumPy in `src/nn_from_scratch.py`, verify gradients analytically, reproduce in PyTorch, and benchmark against Week 2 LightGBM.

- **NumPy Implementation:** Code forward pass, ReLU activation, Softmax output, Cross-Entropy loss, and analytical backpropagation gradients for weights and biases. Verify on XOR or synthetic dataset.
- **Numerical Gradient Check:** Execute two-sided finite difference gradient checking. Confirm that relative error across all weight matrices is strictly below 1e-5.
- **PyTorch Re-implementation:** Re-implement the identical network using `torch.nn.Module`, `torch.optim.AdamW`, and `torch.nn.CrossEntropyLoss` in `notebooks/05_nn_basics.ipynb`. Validate matching loss trajectories.
- **Tabular Comparison & Regularization:** Train the MLP on tabular features. Evaluate against the tuned LightGBM model. Apply at least two regularization techniques (Dropout 0.2 vs 0.5, L2 Weight Decay, Early Stopping with patience=5) and document why MLPs require careful tuning to compete with tree-based models on tabular features.

ACCEPTANCE CRITERIA

✓ NumPy gradient check achieves relative error strictly below 1e-5 across all parameter tensors.
✓ Training and validation loss curves are plotted and interpreted with explicit overfitting diagnosis.
✓ Written technical summary explains the comparative inductive biases between MLPs and gradient boosted trees.

TASK 3.2 Attenomy Cortex Specialized Model (Choose Track A or Track B)

Days 12-14 : Wed-Fri

Select one specialized engineering track aligned with Attenomy Cortex (mentor confirms). Implement transfer learning or fine-tuning, evaluate against classical baselines, and package a single-command training script.

TRACK A: Attenomy CORTEX VISION : COGNITIVE DOCUMENT & INVOICE EXTRACTOR

Power Attenomy Cortex cognitive extraction pipeline by classifying document layouts (invoices, receipts, legal contracts, purchase orders). Fine-tune a pretrained **ResNet18** or **MobileNetV3-Small** using PyTorch `torchvision`. Freeze convolutional backbone, train custom classifier head, then unfreeze top residual block. Apply realistic document augmentations (perspective warp, lighting gradient, color jitter, Gaussian noise). Ensure training runs via `python src/train.py --config configs/vision_config.yaml`.

TRACK B: Attenomy CORTEX NLP & RAG : QUERY INTENT ROUTER & CHUNK RERANKER

Power Attenomy Cortex Enterprise RAG knowledge engine by classifying customer operational intent and computing semantic relevance scores for knowledge retrieval chunks. Build a TF-IDF + Logistic Regression baseline, then fine-tune a small pretrained transformer (**DistilBERT** or **all-MiniLM-L6-v2**) using Hugging Face `transformers`. Benchmark Top-3 retrieval recall, latency, and classification accuracy. Ensure training runs via `python src/train.py --config configs/nlp_config.yaml`.

ACCEPTANCE CRITERIA

✓ Fine-tuned deep learning model outperforms classical baseline, or the performance gap is systematically analyzed.
✓ Train, validation, and test splits are strictly isolated; zero evaluation data used in hyperparameter tuning.
✓ Training is fully reproducible via a single CLI command with hardware requirements, batch size, and seed logged.

TASK 3.3 Disciplined Experiment Tracking & DL Error Analysis

Day 15 : Fri

Track at least 6 systematic deep learning experiments using MLflow or Weights & Biases, conduct diagnostic error analysis, and submit a 1-page proposal for the Week 4 production capstone project for Attenomy Cortex.

- **Experiment Tracking:** Record at least 6 distinct training runs in MLflow or Weights & Biases varying learning rate (1e-4 vs 5e-5), batch size (16 vs 32), dropout (0.2 vs 0.5), and backbone unfreezing schedules.
- **Diagnostic Error Analysis:** Author **`reports/week3_dl_analysis.md`** identifying the 3 most impactful architectural changes and the 3 largest failure modes (e.g. low-resolution camera uploads, rare intent phrasing).
- **Capstone Proposal:** Submit a 1-page proposal (**`reports/capstone_proposal.pdf`**) outlining the proposed Week 4 production system for Attenomy Cortex: problem statement, data pipeline, baseline, advanced architecture, target latency, and operational risks.

ACCEPTANCE CRITERIA

✓ Every logged experiment in MLflow/W&B is reproducible from its recorded configuration parameters.
✓ Diagnostic analysis connects at least 2 observed failure modes to concrete pipeline or data augmentations.
✓ Capstone proposal receives formal sign-off from Attenomy ML Lead during Friday Gate 3 review.

ATTENOMY ENGINEERING STANDARD : EDGE MODEL LATENCY BUDGET

Attenomy Cortex production microservices enforce a strict sub-50ms inference latency budget. MobileNetV3 and DistilBERT are specifically selected over heavy foundation models to guarantee real-time response times on standard cloud CPU instances without requiring costly dedicated GPUs.

03 Week 3 Deliverables & Artifacts Matrix

VERIFICATION MATRIX

ARTIFACT NAME	REPOSITORY TARGET PATH	PRIMARY REQUIREMENT	EVALUATION
NumPy MLP Module	`src/nn_from_scratch.py`	Numerical gradient check error < 1e-5	Score: 10%
Foundations Notebook	`notebooks/05_nn_basics.ipynb`	PyTorch reproduction + LightGBM benchmark	Score: 10%
Training Script	`src/train.py`	Reproducible CLI execution with YAML config	Pass / Fail
Domain Checkpoint	`models/weights/best_weights.pt`	Trained weights (Vision or NLP track)	Score: 10%
Confusion Matrix	`reports/figures/confusion_matrix.png`	Test split evaluation with class metrics	Pass / Fail
DL Analysis Memo	`reports/week3_dl_analysis.md`	6 tracked runs, 3 improvements, 3 failures	Score: 10%
Capstone Proposal	`reports/capstone_proposal.pdf`	1-page proposal approved by mentor	Pass / Fail (Gate)

04 Weekly Submission Deadline & Weekend 1-on-1 Feedback

GATE 3 ASSESSMENT

Weekly Submission Deadline: There is no fixed daily stand-up. Interns work flexibly (15-20 hours/week) and must submit all training scripts, model artifacts, dashboards, and the Capstone proposal to the repository by end of Friday (11:59 PM). On Saturday or Sunday, you will attend an online 1-on-1 meeting for personalized technical feedback, architecture evaluation, and Gate 3 clearance. Passing Gate 3 is mandatory to unlock Week 4 production deployment.

GATE 3 PASSING CONDITION & MENTOR PROBE QUESTIONS

✓ **Mandatory Pass Condition:** NumPy gradient check passes (<1e-5), one working deep learning model with 6+ tracked experiments, and Attenomy Cortex capstone proposal approved.

- **Technical Probe 1:** "Derive the backpropagation gradient for layer 1 weights using the chain rule. Why do vanishing gradients occur in deep networks?"
- **Technical Probe 2:** "Why did you select MobileNetV3 / DistilBERT over larger foundation models? How does parameter size impact deployment latency in Attenomy Cortex?"
- **Technical Probe 3:** "Review your experiment log: what was the single most impactful hyperparameter change, and how did it affect validation loss?"

05 Actionable Daily Execution Checklist

INTERN CHECKLIST

- ☐ **Day 11:** Implement NumPy MLP in src/nn_from_scratch.py, run numerical gradient check.
- ☐ **Day 12:** Port to PyTorch, confirm loss curves match, compare tabular MLP vs LightGBM.
- ☐ **Day 13:** Scaffold Track A (Vision) or Track B (NLP) for Attenomy Cortex, set data augmentations.
- ☐ **Day 14:** Fine-tune transfer model, log at least 6 runs in MLflow/W&B, save weights.
- ☐ **Day 15:** Complete error analysis memo, draft 1-page capstone proposal, submit by end of Friday (11:59 PM), attend weekend review.
- ☐ **Reflect:** Log 3 technical entries in LEARNING_LOG.md with concepts and code links.

06 Flexible Engineering Cadence & Mentorship Support SLA

ENGINEERING CADENCE

Flexible Workflow: No fixed daily stand-ups. Interns manage their hours flexibly to meet the 15-20 hours/week commitment while pushing regular atomic git commits (`feat:`, `fix:`, `test:`).
Weekend Feedback: Every Saturday or Sunday, you will receive personalized 1-on-1 feedback and model reviews from an Attenomy ML Lead.
45-Minute Help Escalation Rule: When blocked by a bug, investigate independently for 45-60 minutes before messaging your mentor with: (1) your objective, (2) the error traceback, and (3) your troubleshooting attempts.